

Engineering Thermodynamics**UNIT 5 Refrigeration cycles and systems****PART A****1. Compare vapour compression system and vapour absorption system. (NOV 2016)**

S.No.	Vapour Compression	Vapour absorption
1	The system has more wear and tear and produce noise due to moving parts the compressor fan etc..	Only moving part in the system is aqua pump. Hence the system is quieter in operation and subject to very little wear.
2	Electric power is needed to drive the system.	Waste or Exhaust steam may be used. No need for electric power.
3	Capacity of system drops rapidly with lowered evaporator pressure.	Capacity of system decreases very little with lowered evaporative pressure by increasing steam pressure to the generator.
4	Mechanical energy is supplied through compressor.	Heat energy is utilized.
5	Charging of refrigerant of the system is easy.	Charging of refrigerant is difficult.

2. Define Tonne of Refrigeration.(MAY 2016, JAN 2016)

The capacity of the refrigeration system is expressed in ton of refrigeration which is the unit of refrigeration.

A ton of refrigeration is defined as the quantity of heat absorbed in order to form one ton of ice in 24 hours when the initial temperature of water is zero degree centigrade.
 1 ton of refrigeration = 210 kJ/min = 3.5 kW

3. Define capacity of refrigerator? (JAN 2015)

It is defined as the rate at which heat can be removed from the cold body. It is called as the capacity of a refrigerator. It is expressed in terms of Tones of Refrigeration.

4. Define reverse Carnot cycle. (NOV 2017)

A reversible cycle is an ideal hypothetical cycle in which all the processes constituting the cycle are reversible. Carnot cycle is a reversible cycle. For a stationary system, as in a piston and cylinder machine, the cycle consists of the following four successive processes

- Reversible isothermal expansion process
- Reversible adiabatic expansion process
- Reversible isothermal compression process
- Reversible adiabatic compression process

5. What is meant by liquefaction? (NOV 2017)

An important area of refrigeration is the liquefaction of gases since many engineering processes at temperature below -100°C (Cryogenic temperature) depends on the liquefied gases. Example: Separation of Oxygen and Nitrogen from air, Preparation of liquid propellants for rockets.

6. State a few properties of an ideal Refrigerant. (MAY 2019, MAY 2016)

- It should be non-flammable.
- It should be non-explosive.
- It should be non-toxic.
- It should not react with moisture to form acids.
- It should not contaminate the food stuff or other stored products.

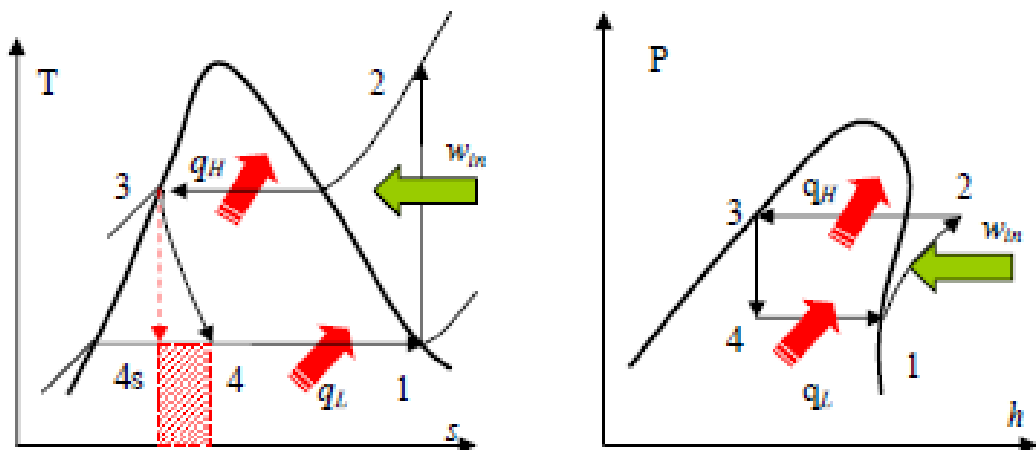
7. Define the term COP. (NOV 2016, MAY 2018)

It is defined as the ratio of desired effect (quantity of heat removed from a cold body) to the net work done on the system

$$\text{COP} = \text{Desired effect} / \text{work input (or)}$$

$$\text{COP} = Q_2 / W$$

8. Draw P-H and T-S diagram of vapour compression refrigerator. (JAN 2015)



T-s and p-h diagrams for an ideal vapor-compression refrigeration cycle

- ✓ Process 1-2 = Adiabatic compression
- ✓ Process 2-3 = Constant Temperature heat rejection
- ✓ Process 3-4 = Adiabatic Expansion
- ✓ Process 4-1 = Constant Temperature heat addition

9. What are the major components of a vapour absorption system? (JAN 2016)

It consists of five main components, namely

- Condenser

- Receiver
- Throttle valve (or) expansion valve
- Evaporator
- Pump
- Analyzer
- Rectifier

10. List out the most widely used refrigerants. (NOV 2018, MAY 2017)

- Ammonia (NH₃).
- Carbon dioxide.(CO₂)
- Sulphur dioxide. (SO₂)
- Freon-12

11. What is meant by refrigeration? (NOV 2018)

Refrigeration is defined as a method of reducing the temperature of a system below that of the surroundings and maintaining it at the lower temperature by continuously extracting the heat from it.

12. What is sub cooling in vapor Compression refrigeration systems? (MAY 2017)

The term subcooling (also called undercooling) refers to a liquid existing at a temperature below its normal boiling point.

It is amount of heat removed from a refrigerants after it has already condensed to liquid.

Liquid refrigerant is sub cooled at constant pressure before entering to expansion valve.

13. Define performance factor of a Refrigeration system. (MAY 2019)

Co efficient of performance is ratio of between desired effects to the work input

$$\text{COP} = \text{Desired effect} / \text{work input (or)}$$

$$\text{COP} = Q_2 / W$$

14. List the components of vapour compression refrigeration system. (MAY 2018)

It consists of five main components, namely

- Compressor
- Condenser
- Receiver
- Throttle valve (or) expansion valve
- Evaporator

Part B

1. With a suitable sketch explain the basic operations involved in a vapour compression refrigeration plant.(Nov 17, May 16,Jan 15,16,Apr 17)

Vapour compression Refrigeration system uses a liquefiable vapour as the refrigerant. In this system, the refrigerant used alternatively undergoes a change of phase from vapour to liquid and liquid to vapour during the cycle. In this cycle, the isentropic expansion of Carnot cycle is replaced by a throttling process, i.e. expansion in a valve or capillary tube. A simple vapour compression refrigeration system is shown in Figure 5.1.

The latent heat of vaporization is utilized for absorbing the heat from low temperature space and it transfers to surrounding so that the space is maintained at a temperature lower than that of surrounding

The main parts are

1. **Compressor.** The low pressure and temperature vapor refrigerant from evaporation is drawn into the compressor through the inlet or suction valve A, where it is compressed to a high pressure and temperature. The high pressure and temperature vapor refrigerant is discharged into the condenser through the delivery or discharge valve B.
2. **Condenser.** The condenser or cooler consists of coils of pipe in which the high pressure and temperature vapor refrigerant is cooled and condensed. The refrigerant, while passing through the condenser, gives up its latent heat to the surrounding condensing medium which is normally air or water.
3. **Receiver.** The condensed liquid refrigerant from the condenser is stored in a vessel known as receiver form where it is supplied to the evaporator through the expansion valve or refrigerant control valve.
4. **Expansion valve.** It is also called throttle valve or refrigerant control valve. The function of the expansion valve is to allow the liquid refrigerant under high pressure and temperature to pass at a controlled rate after reducing its pressure and temperature. Some of the liquid refrigerant evaporates as it passes through the expansion valve, but the greater portion is vaporized in the evaporator at the low pressure and temperature.
5. **Evaporator.** An evaporator consists of coils of pipe in which the liquid-vapour refrigerant at low pressure and temperature is evaporated and changed into vapour refrigerant at low pressure and temperature. In evaporating, the liquid vapour refrigerant absorbs its latent heat of vaporization from the medium (air, water or brine) which is to be cooled.

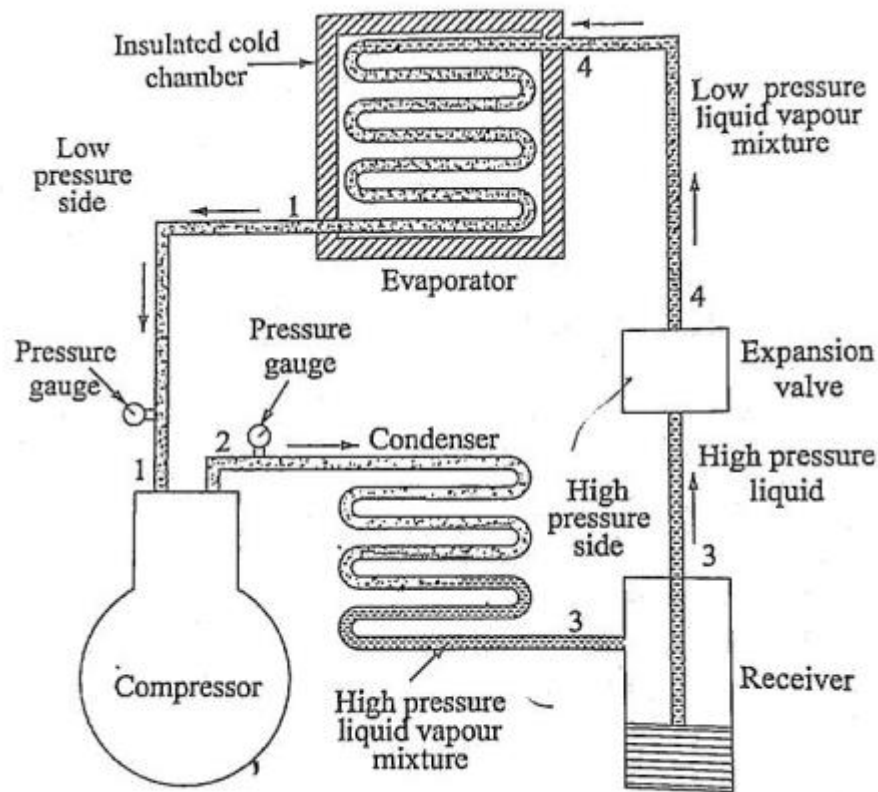


Figure 5.1 Vapour compression system

Working:

The high pressure vapour refrigerant from the condenser enters the expansion or throttle valve where it expands to the required pressure. During expansion, the pressure is reduced and the vapour becomes partially converted into liquid and produces the cooling effect. This process is represented in p-h diagram as 3-4.

A mixture of low temperature vapour and liquid from expansion valve is then enters the evaporator and absorbs heat from the space to be refrigerated. Thus, the "Refrigerating effect" is obtained. This process is represented in p-h diagram as 4-1. Because of this, the liquid becomes vapour (state 1) and this vapour enters the compressor. There it is compressed, thus, the pressure goes up. This process is represented in p-h diagram as 1-2. This high pressure vapour enters the condenser. There, it loses its latent heat and becomes liquid (Process 5). The liquid refrigerant is stored in a receiver and supplied to the expansion valve, and thus, the cycle is continued.

P-h diagram:

The above processes are represented on a p - h and T-s chart as shown in Figure 5.2(a) and Figure 5.2 (b) respectively

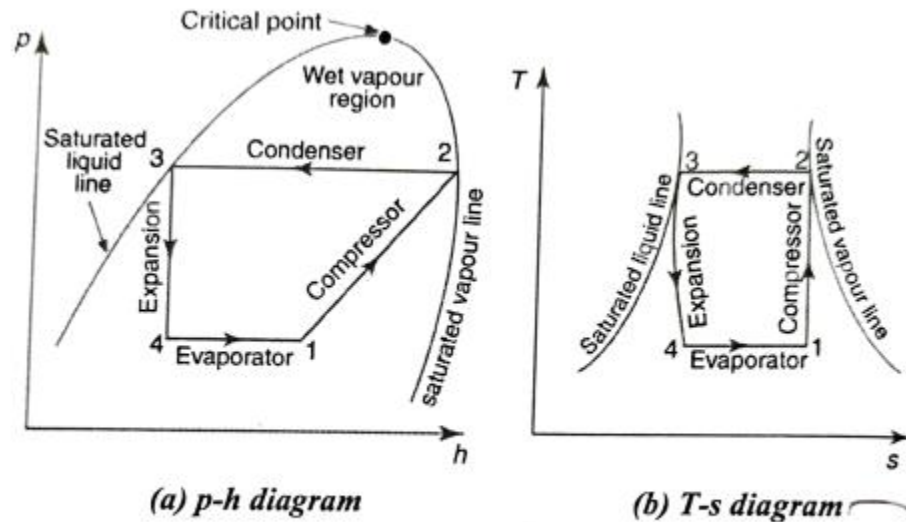
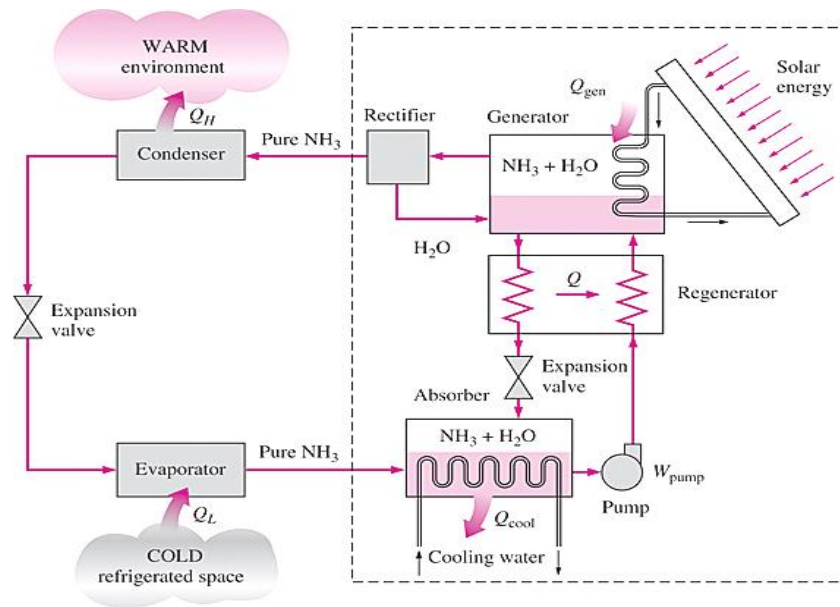


Figure 5.2 Vapour compression refrigeration on p - h plot

2. Explain the working principle of vapour absorption refrigeration cycle with neat sketch. (Nov 16, Jan 15, Apr 17)

- The vapour absorption refrigeration system is a heat operated system.
- It differs from the VCRS only in the manner in which the circulation of the refrigerant is achieved. In this system, the compressor is replaced by an absorber, a generator and a pump.
- The refrigerant used in this system must be highly soluble in the solution known as absorbent.
- The most widely used system is the ammonia–water system, where ammonia (NH_3) serves as the refrigerant and water (H_2O) as the transport medium.
- Other systems include water–lithium bromide and water–lithium chloride systems, where water serves as the refrigerant. These systems are limited to applications such as A-C where the minimum temperature is above the freezing point of water.
- The liquid refrigerant in the evaporator absorbs the heat from the medium to be cooled and it undergoes a change of phase from liquid to vapour.
- The low pressure vapour is then passed to the absorber. In the absorber, the low pressure ammonia vapour is dissolved in the weak ammonia solution producing strong ammonia solution at low pressure.



Schematic for vapour absorption refrigeration system.

- The strong ammonia solution is then pumped to a generator through the heat exchanger at high pressure.
- While passing through the heat exchanger, the strong ammonia solution is warmed up by the hot weak ammonia solution flowing from the generator to absorber.
- The warm strong ammonia solution is heated by an external source in the generator. Some examples of external sources include geothermal energy, solar energy, and waste heat from cogeneration or process steam plants, and even natural gas when it is at a relatively low price.
- Due to the heating, the vapour gets separated from the solution.
- The vapour which is at high pressure and high temperature is condensed to low temperature in the condenser by cold water circulation.
- The high pressure liquid ammonia then passes through the expansion valve where it is expanded to low pressure and temperature.
- The low pressure and low temperature ammonia liquid again enters the evaporator where it absorbs the heat from the medium to be cooled and the cycle continues.

3. Explain with a neat sketch the principle and operation of a lithium-Bromide absorption system.(May 19)

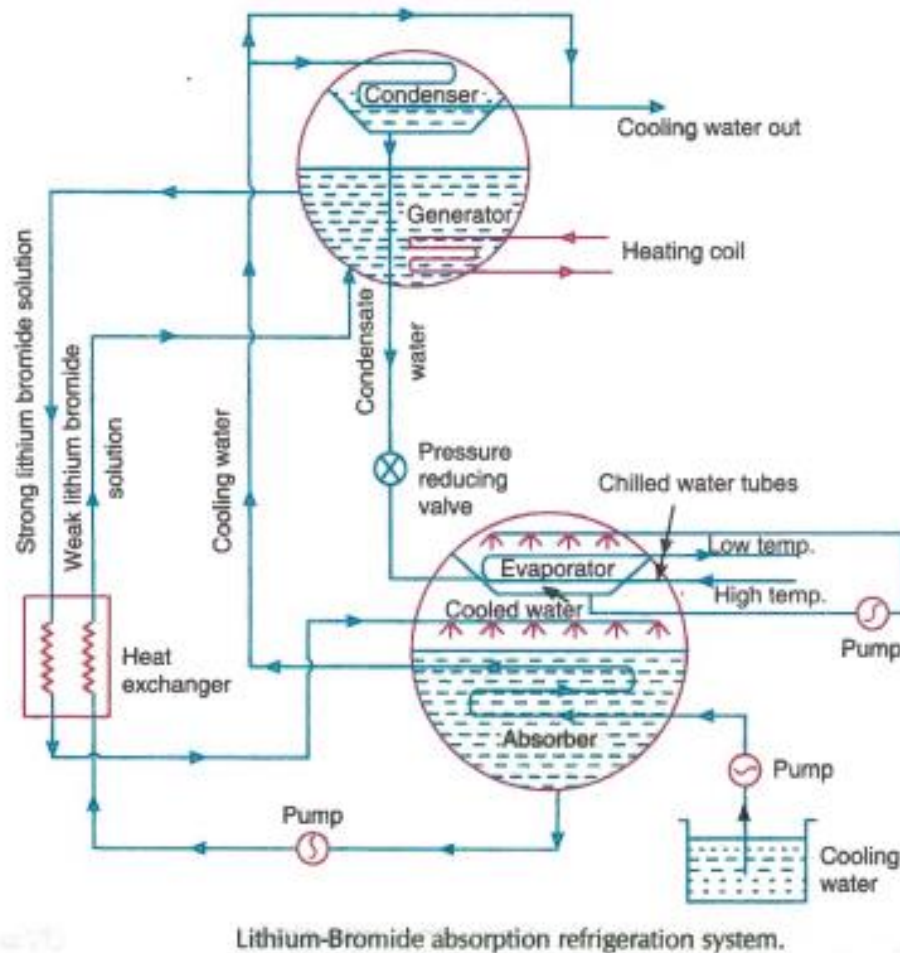
The lithium bromide absorption refrigeration system uses a solution of lithium bromide in water. In this system, the water is being used as a refrigerant whereas lithium bromide, which is a highly hygroscopic salt, as an absorbent. The lithium bromide solution has a strong affinity for water vapour because of its very low vapour pressure. Since lithium bromide solution is corrosive, therefore, inhibitors should be added in order to protect the metal parts of the system against corrosion. Lithium chromate is often used as a corrosion inhibitor. This system is very popular for air-conditioning in which low refrigeration temperatures (not below 0°C) are required.

Fig. 2.5 shows a lithium bromide vapour absorption system. In this system, the absorber and the evaporator are placed in one shell which operates at the same low pressure of the systems. The generator and condenser are placed in another shell which operates at the same high pressure of the system. The principle of operations of this system is discussed below:

- Lithium bromide system is another type of absorption refrigeration uses water as refrigerant and lithium bromide as absorbent. Working of this system is shown in Figure 5.20.
- This system is mainly used for water-chilling purposes.
- Water to be chilled is passed through tubes. Refrigerant water, after expanding in the expansion valve is sprayed over these tubes.
- Because of this, the tube water gets chilled and the refrigerant water becomes water vapour
- Lithium bromide solution, which is sprayed in the absorber, absorbs the water vapour. Thus, the solution becomes weak.
- This weak solution is pumped to the generator. Here, heating of the solution is done. Because of this, some water evaporates. So, the solution becomes strong.
- This strong solution is passed through the heat exchanger and sprayed in the absorber. Thus the cycle repeats.
- Water vapours formed in the generator is passed through a condenser.

Water vapour thus gets condensed and becomes liquid water.

- External supply of cooling water is effected the condenser.
- This cooling water is supplied through the absorber, to carry some heat from there.
- The condensed water (from the condenser) is supplied to the evaporator to make up for the water vapour formed there.

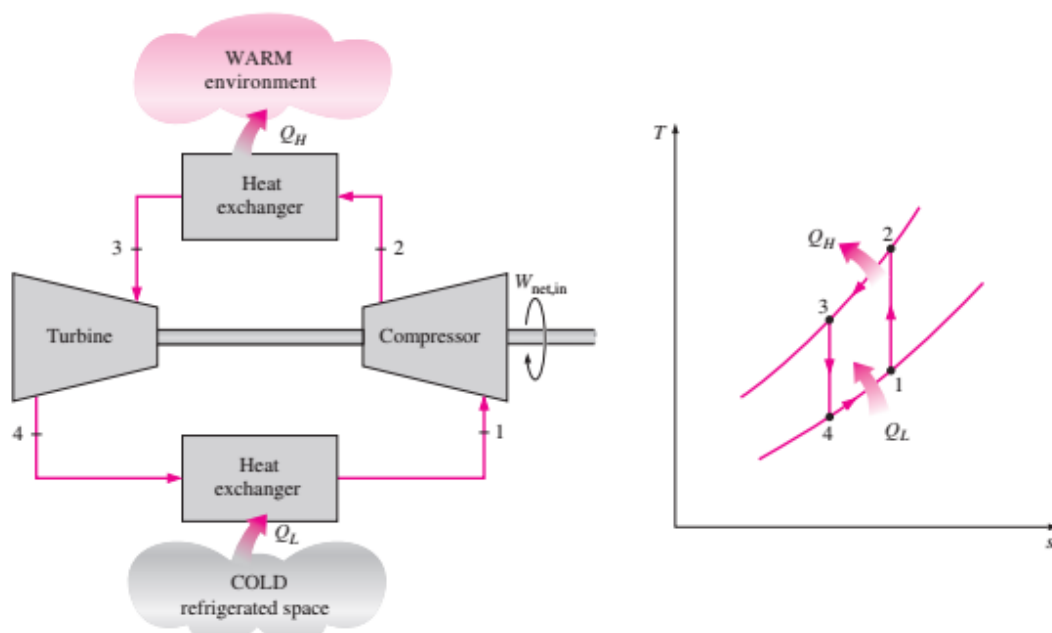


4. Write a note on Gas Refrigeration cycle.(Nov 17)

The Carnot cycle (the standard of comparison for power cycles) and the reversed Carnot cycle (the standard of comparison for refrigeration cycles) are identical, except that the reversed Carnot cycle operates in the reverse direction. This suggests that the power cycles discussed in earlier chapters can be used as refrigeration cycles by simply reversing them. In fact, the vapor-compression refrigeration cycle is essentially a modified Rankine cycle operating in reverse. Another example is the reversed Stirling cycle, which is the cycle on which

Stirling refrigerators operate. In this section, we discuss the *reversed Brayton cycle*, better known as the **gas refrigeration cycle**

Consider the gas refrigeration cycle shown in Fig. The surroundings are at T_0 , and the refrigerated space is to be maintained at T_L . The gas is compressed during process 1-2. The high-pressure, high-temperature gas at state 2 is then cooled at constant pressure to T_0 by rejecting heat to the surroundings. This is followed by an expansion process in a turbine, during which the gas temperature drops to T_4 . Finally, the cool gas absorbs heat from the refrigerated space until its temperature rises to T_1 .



All the processes described are internally reversible, and the cycle executed is the ideal gas refrigeration cycle. In actual gas refrigeration cycles, the compression and expansion processes deviate from the isentropic ones, and T_3 is higher than T_0 unless the heat exchanger is infinitely large. On a $T-s$ diagram, the area under process curve 4-1 represents the heat removed from the refrigerated space, and the enclosed area 1-2-3-4-1 represents the net work input. The ratio of these areas is the COP for the cycle, which may be expressed as

$$\text{COP}_R = \frac{q_L}{W_{\text{net,in}}} = \frac{q_L}{W_{\text{comp,in}} - W_{\text{turb,out}}}$$

$$q_L = h_1 - h_4$$

$$W_{\text{turb,out}} = h_3 - h_4$$

$$W_{\text{comp,in}} = h_2 - h_1$$

5. Write a note on COP. (Nov 17)

Co-Efficient Of Performance (COP):

- Co efficient of performance is ratio between desired effects to the work input
- The COP characterizes the amount of heat removed from cooler.
- The COP for cooling mode is less than for heating mode, because the work done by compressor is utilized only during the heating mode.
- The operating principle of refrigerators, air conditioners, and heat pumps is the same and it is just the reverse of a heat engine.
- In simple words, heat pump consumes work (electric energy) and removes heat from one side and gives this heat to the other side.
- From an economic point of view, the best refrigeration cycle is one that removes the greatest amount of heat from the inside of the refrigerator (cold reservoir) for the least expenditure of mechanical work or electric energy.
- The COP strongly depends on outside temperature and required indoor temperature. For temperature difference of about 25°C (45 – 20) the COP may be about 2.5, while for difference of about 8 °C (30 – 22) the COP may reach 3.5.

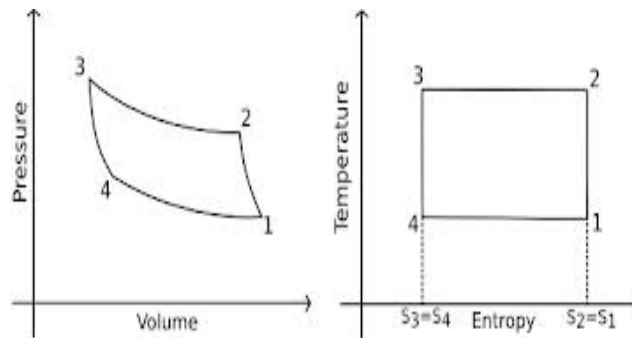
$$\text{COP} = \text{Desired effect} / \text{work input (or)}$$

$$\text{COP} = Q_2 / W$$

6. Explain the Reversed Carnot Cycle with P-V and T-S diagram.

- Reversing the Carnot cycle does reverse the directions of heat and work interactions.
- A refrigerator or heat pump that operates on the reversed Carnot cycle is called a Carnot refrigerator or a Carnot heat pump.
- Reversed Carnot cycle is shown in Fig.6.1. It consists of the following processes.
- Process 1-2: Isentropic compression of the working fluid with the aid of external work. The temperature of the fluid rises from T_2 to T_1 .

- Process 2-3: Isothermal compression of the working fluid during which heat is rejected at constant high temperature T_1 .
- Process 3-4: Isentropic expansion of the working fluid. The temperature of the working fluid falls from T_1 to T_2 .
- Process 4-1: Absorption of heat by the working fluid from refrigerator at constant low temperature T_2 during isothermal expansion.



Reversed Carnot Cycle

COP of Refrigerator

$(\text{COP})_{\text{ref}} = \text{Refrigerating effect/work done}$

$$(\text{COP})_{\text{ref}} = Q_e / (Q_r - Q_e) = T_1 / (T_2 - T_1)$$

- Practically, the reversed Carnot cycle cannot be used for refrigeration purpose as the isentropic process requires very high speed operation, whereas the isothermal process requires very low speed operation.

ADDITIONAL QUESTIONS

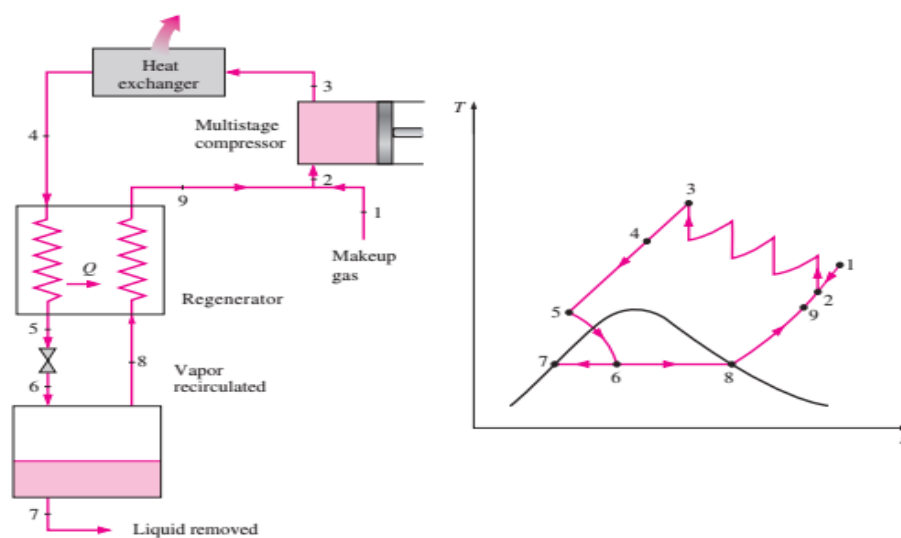
7. Explain the Liquefaction of Gases.

The liquefaction of gases has always been an important area of refrigeration since many important scientific and engineering processes at cryogenic temperatures (temperatures below about 100°C) depend on liquefied gases. Some examples of such processes are the separation of oxygen and nitrogen from air, preparation of liquid propellants for rockets, the study of material properties at low temperatures, and the study of some exciting phenomena such as superconductivity.

At temperatures above the critical-point value, a substance exists in the gas phase only. The critical temperatures of helium, hydrogen, and nitrogen (three commonly used liquefied gases) are 268, 240, and 147°C, respectively. Therefore, none of these substances exist in liquid form at atmospheric conditions. Furthermore, low temperatures of this magnitude cannot be obtained by ordinary refrigeration techniques.

Several cycles, some complex and others simple, are used successfully for the liquefaction of gases. Below we discuss the Linde-Hampson cycle, which is shown schematically and on a T - s diagram in Fig.

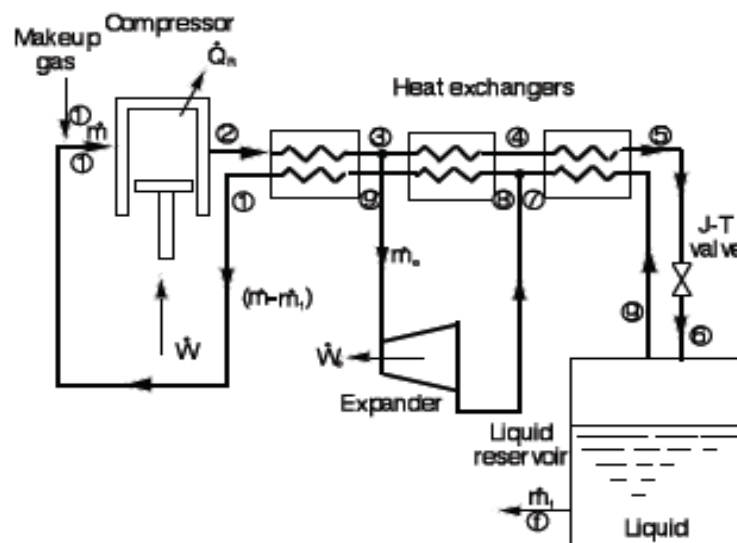
Makeup gas is mixed with the uncondensed portion of the gas from the previous cycle, and the mixture at state 2 is compressed by a multistage compressor to state 3. The compression process approaches an isothermal process due to intercooling. The high-pressure gas is cooled in an aftercooler by a cooling medium or by a separate external refrigeration system to state 4. The gas is further cooled in a regenerative counter-flow heat exchanger by the uncondensed portion of gas from the previous cycle to state 5, and it is throttled to state 6, which is a saturated liquid–vapor mixture state. The liquid (state 7) is collected as the desired product, and the vapor (state 8) is routed through the regenerator to cool the high-pressure gas approaching the throttling valve. Finally, the gas is mixed with fresh makeup gas, and the cycle is repeated.



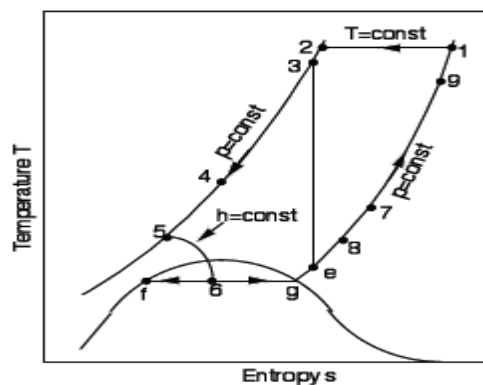
Linde-Hampson system for liquefying gases.

8. Explain with neat sketch the working of Claude System for Liquefaction of Air.

- In Claude system, energy is removed from the gas stream by allowing it to do some work in an expander.
- The flow and T-s diagrams are given in Figure 5.9 and 5.10.
- The gas is first compressed to pressures of about 40atm and then passed through the first heat exchanger.
- Approximately 80% of the gas is then diverted from the main stream, expanded through an expander, and reunited with the return stream below the second heat exchanger.
- The stream to be liquefied continues through the second and third heat exchangers, and is finally expanded through an expansion valve to the liquid receiver.
- The cold vapour from the liquid receiver is returned through the heat exchangers to cool the incoming gas.



Schematic diagram for Claude System of air liquefaction cycle



Temperature-entropy diagram for Claude System of Air Liquefaction

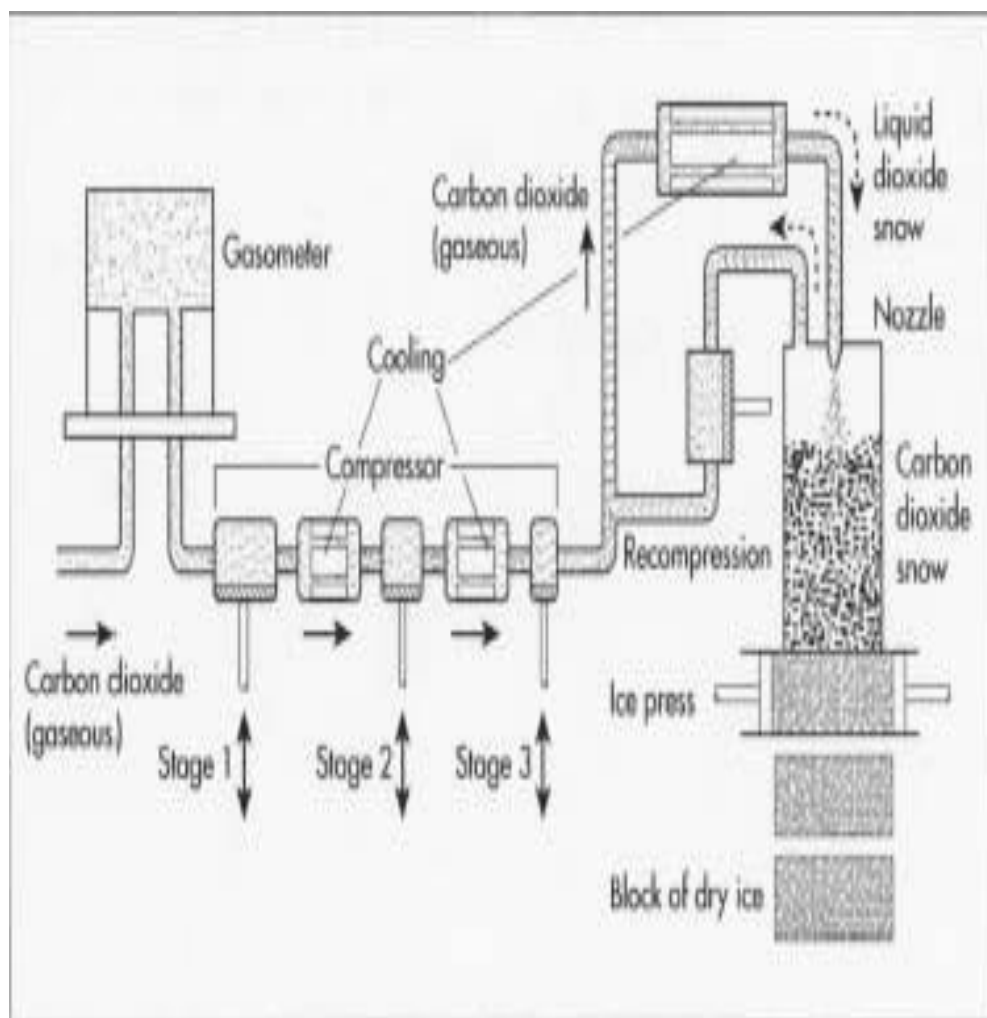
9. Explain the details about Production of Solid Ice.

- Dry ice is the name given to carbon dioxide when it is in a solid state.
- Carbon dioxide is found in the earth's atmosphere; it is a gas that humans exhale and plants use for photosynthesis.
- This chemical compound is colorless, odorless, tasteless, and about 1.5 times as dense as air.
- Carbon dioxide turns from gas to an opaque white solid while under pressure and at low temperatures, turning solid at -109°F (178.5°C).
- Dry ice is manufactured primarily in two forms, either as a block of dry ice which weighs over 50 lb (22.7 kg) or in small pieces that vary in size from the size of a grain of rice to a larger pellet.
- Dry ice does not melt, instead it sublimates, meaning the solid turns directly into a gas (bypassing the liquid state) as the temperature rises and the solid begins to dissipate.

Manufacturing Process of Dry Ice

- Carbon dioxide is liquefied by compressing and cooling, liquefying at a pressure of approximately 870 lb/in² (395 kg/cm²) at room temperature. Liquid carbon dioxide is pumped, via piping, into huge holding tanks so that dry ice manufacturers can remove the liquid required.
- The pressurized, refrigerated liquid carbon dioxide is piped directly into a pressurized tank or rail car owned by the dry ice manufacturer and heads for the plant.
- The tank truck pulls up to the factory and dumps the liquid carbon dioxide into huge tanks on the premises.
- These tanks hold the liquid under pressure, keeping it refrigerated so that it remains in liquid state.
- These tanks are situated adjacent to the factory wall and, through piping, the liquid is brought directly inside when required for manufacturing.
- The liquid carbon dioxide is released, again via piping, from the adjacent tanks through the factory wall and into the dry ice press.
- When the liquid moves from a highly-pressurized environment to atmospheric pressure, it expands and evaporates at high speeds, causing the liquid to cool to its freezing point which is -109°F (-78.3°C).

- A nozzle puts the liquid into the top block of a dry ice press, which stands approximately 16 ft (4.9 m) tall.
- This press includes a large block at the top that can exert extreme pressure on the product that is brought into it.
- When the liquid carbon dioxide hits the block of the dry ice press, it immediately solidifies since it is now at room temperature. The carbon dioxide now resembles snow.
- This snow, now in the upper portion of the press, must be compressed into a block of dry ice.
- Thus, this top portion of the press goes up and down with extraordinary pressure (about 60 tons), squashing the snow into a solid block of dry ice. This is approximately a five minute process.
- When the block is solid, it is generally about 2 ft (61 cm) wide and 10 in (25 cm) high, weighing about 220 lb (100 kg)



The formation of dry ice is a series of chemical reactions.

